IEOR 221 - HW 3 - Jie Wen

1

The interest rate swap has a notional principal of $100 million and exchanges six-month LIBOR for a fixed rate of 4% per annum (compounded semiannually), implying fixed payments of $2 million every six months. The remaining life is 10 months, with payments due in 4 months and 10 months. The six-month LIBOR rate fixed two months ago is 2.4%, so the floating payment in 4 months is $1.2 million. The six-month LIBOR forward rate for the period starting in 4 months is 3%, so the expected floating payment in 10 months is $1.5 million. All OIS rates are 2.7% with continuous compounding, so the discount factors are $e^{-0.027\times(4/12)} = e^{-0.009} \approx 0.99104$ for 4 months and $e^{-0.027\times(10/12)} = e^{-0.0225} \approx 0.97775$ for 10 months.

The value of the swap is the present value of the net cash flows. To the party paying floating (receiving fixed), the value is $(2-1.2)\times 10^6 \times 0.99104 + (2-1.5)\times 10^6 \times 0.97775 \approx 1,281,708$. To the party paying fixed (receiving floating), the value is $-1,281,708$.

2

The currency swap involves annual fixed interest payments and a principal exchange at maturity in 15 months (1.25 years). Consistent with standard valuation for such swaps with non-integer remaining terms, we assume the next interest payment is in 3 months (0.25 years) and the final interest payment plus principal exchange is at maturity in 1.25 years.

The party paying sterling pays 10% annually on £20 million (£2 million per payment) and receives 6% annually on $30 million ($1.8 million per payment). At maturity, this party also pays £20 million principal and receives $30 million principal.

Cash flows for the party paying sterling:

- **At 0.25 years:** Pay £2 million, receive $1.8 million.
- **At 1.25 years:** Pay £22 million (£2 million interest + £20 million principal), receive $31.8 million ($1.8 million interest + $30 million principal).

Valuation of the Swap in USD:

- **Compute the present value (PV) of the USD cash flows received, discounted at the US risk-free rate of 4% (annual compounding):**
  - Discount Factor at 0.25 years: $DF_{USD}(0.25) = \frac{1}{(1.04)^{0.25}} \approx 0.9902$
  - Discount Factor at 1.25 years: $DF_{USD}(1.25) = \frac{1}{(1.04)^{1.25}} \approx 0.9522$
  - Present Value of USD cash flows:

$$PV_{USD_received} \approx (0.9902 \times 1,800,000) + (0.9522 \times 31,800,000) \approx \$32,061,013$$

- **Compute the PV of the GBP cash flows paid, discounted at the UK risk-free rate of 7% (annual compounding):**
  - Discount Factor at 0.25 years: $DF_{GBP}(0.25) = \frac{1}{(1.07)^{0.25}} \approx 0.9832$
  - Discount Factor at 1.25 years: $DF_{GBP}(1.25) = \frac{1}{(1.07)^{1.25}} \approx 0.9189$
  - Present Value of GBP cash flows:

$$PV_{GBP_paid} \approx (0.9832 \times 2,000,000) + (0.9189 \times 22,000,000) \approx £22,182,349.5$$

- **Convert the GBP PV to USD using the current exchange rate of 1.5500 USD/GBP:**

$$PV_{paid_USD} \approx 1.5500 \times 22,182,349.5 \approx \$34,382,642$$

- **Value to the party paying sterling = PV of cash received - PV of cash paid:**

$$\text{Value} \approx \$32,061,013 - \$34,382,642 = -\$2,321,629$$

The value to the party paying dollars is the negative of this amount: **+$2,321,629**.

3

Six-month LIBOR was 3% per annum halfway through year 3 (at year 2.5). Therefore, the floating payment due at the default date (year 3) is based on LIBOR = 3%.

- Fixed leg payment (received): $10M × 4% × 0.5 = $200,000
- Floating leg payment (paid): $10M × 3% × 0.5 = $150,000

Net payment at default: $200,000 - $150,000 = $50,000 (received by financial institution)

At default (year 3):

Forward Rates and Discount Factors

- Six-month forward LIBOR rates for all maturities: 2% per annum (semiannual compounding)
- OIS rates: 1.8% per annum (continuous compounding)

Discount Factors using OIS rates:

- Year 3.5 (0.5 years from default): $DF = e^{-0.018 \times 0.5} \approx 0.99104$
- Year 4.0 (1.0 years from default): $DF = e^{-0.018 \times 1.0} \approx 0.98216$
- Year 4.5 (1.5 years from default): $DF = e^{-0.018 \times 1.5} \approx 0.97336$
- Year 5.0 (2.0 years from default): $DF = e^{-0.018 \times 2.0} \approx 0.96464$

Calculate Remaining Cash Flows

Each period:

- Fixed received: $10M × 4% × 0.5 = $200,000
- Floating paid: $10M × 2% × 0.5 = $100,000
- Net cash flow per period: $200,000 - $100,000 = $100,000

Present Value of Remaining Swap

Period	Years from Default	Net Cash Flow	Discount Factor	Present Value
3.5	0.5	$100,000	0.99104	$99,104
4.0	1.0	$100,000	0.98216	$98,216
4.5	1.5	$100,000	0.97336	$97,336
5.0	2.0	$100,000	0.96464	$96,464
Total				**$391,120**

Total Loss = $50,000 + $391,120 = $441,120

This loss occurs because the financial institution was receiving a favorable fixed rate of 4% while floating rates have declined to 2%, making the remaining swap payments valuable, but this value is lost due to the counterparty default.

4

This is an interest rate swap (receiving fixed, paying floating) valued as $V = N \sum_{i=1}^{n} (R_S - R_F^{i-1,i}) \delta_i DF_i$, where $N = \$100,000,000$ is the notional, $R_S = 0.03$ is the fixed rate received, $R_F^{i-1,i}$ is the forward rate for period i, $\delta_i = 1$ is the time fraction (annual payments), and DF_i is the discount factor to time i (in years).

The OIS curve is flat at 3.4% with annual compounding, so the zero rate is 3.4% for all maturities and all forward rates $R_F = 0.034$. The discount factors are:

- $DF_1 = \frac{1}{1+0.034}$
- $DF_2 = \frac{1}{(1+0.034)^2}$

Thus,

$$V = 100,000,000 \times (0.03 - 0.034) \times (DF_1 + DF_2) \approx -760,974$$